Code No: 117CF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, December-2023/January-2024
DESIGN PATTERNS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Define Design Pattern and explain.	[2]
b)	How to select a design pattern? Explain.	[3]
c)	What are the consequences of Builder pattern?	[2]
d)	Discuss about the problems associated in designing a document editor?	[3]
e)	Write the structure of Façade pattern.	[2]
f)	What are the variations of the Adapter Pattern?	[3]
g)	What are the consequences of Observer pattern?	[2]
h)	Describe the purpose of mediator patterns.	[3]
i)	What is the intent of strategy pattern?	[2]
j)	Explain about Pattern Community.	[3]

PART – B

(50 Marks)

2.a) Explain about design patterns in small talk MVC.
b) Describe the structure of the design pattern template. [5+5]

OR

3.a) What are the similarities and dissimilarities of patterns and frameworks? Explain.
b) What is the basis for classifying design patterns? Discuss. [5+5]

4.a) Discuss about the functionality of operations that are accessed indirectly through user operations.
b) Write the sample code of prototype pattern and Singleton patterns. [5+5]

OR

5.a) Discuss the implementation issues of Builder design pattern.
b) Explain about supporting multiple window systems in designing a document Editor.[5+5]

6.a) Write the sample code of Flyweight pattern and explain.
b) What are the functions of the participants of Adapter pattern? Explain. [5+5]

OR

7.a) Explain about the intent, uses and related patterns of Proxy patterns.
b) Discuss the implementation issues of Decorator Pattern. [5+5]

8.a) Discuss about the structure and participants of Interpreter pattern.
b) What are the consequences of Chain of Responsibility pattern? Explain. [5+5]

OR

9.a) Explain about the applicability and structure of Command pattern.
b) Discuss about the structure and participants of Iterator pattern. [5+5]

10.a) Describe the features of behavioral patterns in detail.
b) Explain what is to be expected from design patterns. [5+5]

OR

11.a) Explain about the intent, motivation, structure and applicability of State pattern.
b) Write about the participants, collaborations and sample code of Visitor pattern. [5+5]

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